

# No-Go Theorems for One-Step Lyapunov Potentials for the $3x + 1$ Map

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## Abstract

A natural strategy for the  $3x + 1$  (Collatz) problem is to exhibit a potential  $\Phi(x) = \log_2 x + g(x)$  that is nonincreasing along the accelerated odd map  $f(x) = (3x + 1)/2^{v_2(3x+1)}$ , with a correction  $g$  computed from simple features of  $x$ . It is folklore that no such monotone invariant is known; we prove that for broad natural classes of corrections none can exist. The master mechanism is *shadowing an expanding rational cycle*: the cycle  $-5 \mapsto -7 \mapsto -5$  has multiplier  $3^2/2^3 > 1$ , and its positive-integer shadows  $x \equiv -5 \pmod{2^N}$  traverse its coordinate itinerary while gaining value, so any correction that cannot separate  $x$  from  $f^2(x) = \frac{9}{8}x + \frac{5}{8}$  is refuted by two integers — the smallest certificate being  $1275 \mapsto 1913 \mapsto 1435$ . This excludes all corrections  $g(x \bmod 2^m, \tau(x), \text{len}(x), \lambda_1(x), \dots, \lambda_d(x))$ , where  $\tau$  is the trailing-ones count and the  $\lambda_i$  are suffix-pattern detectors of arbitrary depth (Theorem A). Iterating the shadow and applying Dirichlet approximation to  $\log_2 \frac{9}{8}$  extends the exclusion to corrections using any fixed number of leading binary digits (Theorem B); a Mersenne burn argument excludes all bounded corrections of arbitrary dependence (Theorem C). The detectors are not ad hoc: they read a canonical structure, an exact ancestry tower  $w_d(M) = (2^{M+2d} - 2^{2d+1} + 3^d)/3^d$  with  $f^d(w_d(M)) = 2^M - 1$ , supported on  $M \equiv 1 \pmod{2 \cdot 3^{d-1}}$  with a carry-free periodic binary normal form (Theorem D). The surviving one-step class reads the quantized full logarithm, where further no-go progress becomes conjecture-adjacent. The same cycle also blocks a second method: on the Mersenne spine  $2^n - 1 \mapsto (3^n - 1)/2$ , no finite set of residue-class certificates on  $n$  can force a positive block score, because a 2-adic index  $n^*$  with  $3^{n^*+3} = -367$  sends the orbit into  $-5 \mapsto -7$  (Theorem E).

## 1 Introduction

For odd  $x \in \mathbb{Z}_{>0}$  let

$$f(x) = \frac{3x + 1}{2^{v_2(3x+1)}},$$

the accelerated (odd-to-odd) form of the Collatz map; the  $3x + 1$  conjecture asserts that every orbit reaches 1. Among the strategies proposed for the problem, one of the most natural is the *potential-function* (Lyapunov) method: find  $\Phi : 2\mathbb{Z} + 1 \rightarrow \mathbb{R}$  with  $\Phi(f(x)) \leq \Phi(x)$  for all odd  $x > 1$ , ideally of the coercive form  $\Phi(x) = \log_2 x + g(x)$  with  $g$  a correction of lower order, so that monotonicity constrains orbit growth. Heuristics support the idea: a “random” odd

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\*Repository: [github.com/docbgm2002/collatz-things](https://github.com/docbgm2002/collatz-things). Every numbered claim is machine-verified in exact integer or rational arithmetic; see Section 10.

step multiplies  $x$  by  $3/2^e$  with  $e$  geometrically distributed of mean 2, so  $\log_2 x$  decreases by  $\log_2 \frac{4}{3}$  on average [5]. The difficulty, as surveys note, is that no correction making the decrease pointwise has ever been found [6]. To the best of our knowledge the literature contains no theorem *excluding* a concrete class of such potentials; the absence is reported as an empirical fact. This paper proves exclusions, and locates the precise boundary at which the exclusion program must stop.

Throughout,  $\tau(x) = v_2(x+1)$  is the number of trailing 1s in the binary expansion of odd  $x$  (the *fuel*: Lemma 2.3 shows an odd step converts one unit of fuel into a multiplication by almost exactly  $\frac{3}{2}$ );  $\text{len}(x)$  is the bit length;  $\text{LD}_j(x) = \lfloor x/2^{\text{len}(x)-j} \rfloor$  is the top  $j$  bits; and  $\lambda_1, \lambda_2, \dots$  are the suffix-pattern detectors of Definition 2.5, which read the run length of a canonical hierarchy of periodic binary patterns (levels 1 and 2 being  $(10)^k 1001$  and  $(111000)^k 01$ ). Our main impossibility results:

*Theorem A* (master no-go; Theorem 3.2). For every  $m, d \geq 0$  and every function  $g$ , the potential  $\log_2 x + g(x \bmod 2^m, \tau(x), \text{len}(x), \lambda_1(x), \dots, \lambda_d(x))$  is not nonincreasing along  $f$  on odd  $x > 1$ .

*Theorem B* (leading digits; Theorem 4.1). For every  $m, j, d \geq 0$  and every  $g$ , the potential  $\log_2 x + g(x \bmod 2^m, \tau(x), \text{LD}_j(x), \lambda_1(x), \dots, \lambda_d(x))$  is not nonincreasing along  $f$  on odd  $x > 1$ .

*Theorem C* (bounded corrections; Theorem 5.1). For every bounded  $G : 2\mathbb{Z}+1 \rightarrow \mathbb{R}$ , of arbitrary dependence on  $x$ , the potential  $\log_2 x + G(x)$  is not nonincreasing along  $f$  on odd  $x > 1$ .

Theorem A is proved by two integers per parameter choice: the map has the expanding rational cycle  $-5 \mapsto -7 \mapsto -5$ , of multiplier  $3^2/2^3 = \frac{9}{8} > 1$  (expansion is exactly why it lives at negative integers), and its shadow at positive  $x \equiv -5 \pmod{2^N}$  reproduces the cycle's coordinate itinerary while the value gains a factor  $> \frac{9}{8}$  per period. Any correction blind to the difference between  $x$  and  $f^2(x) = \frac{9}{8}x + \frac{5}{8}$  is refuted outright; the smallest certificate is  $1275 \mapsto 1913 \mapsto 1435$ . Theorem B iterates the shadow  $a$  times and chooses  $a$ , by Dirichlet approximation to the irrational  $\log_2 \frac{9}{8}$ , so that the accumulated gain sits next to an exact power of 2 and the leading digits return. Theorem C is a one-paragraph consequence of the Mersenne burn.

The detectors quantified over in Theorems A and B are given meaning by a structural result of independent interest:

*Theorem D* (Mersenne ancestry tower; Theorem 6.1). For  $d \geq 1$  and  $M \equiv 1 \pmod{2 \cdot 3^{d-1}}$ ,  $M > 1$ , the integer  $w_d(M) = (2^{M+2d} - 2^{2d+1} + 3^d)/3^d$  is odd with  $\tau = 1$ , satisfies  $3w_d(M) + 1 = 4w_{d-1}(M)$  and hence  $f^d(w_d(M)) = 2^M - 1$  with every step of value ratio in  $(\frac{3}{4}, 1)$ ; its binary expansion is exactly periodic (carry-free) of period  $2 \cdot 3^{d-1}$ ; and  $w_d(M) \equiv 1 - 2^{2d+1}3^{-d} \pmod{2^{M-1}}$ , distinct levels separating 2-adically at bit  $2\min(d, d') + 1$  exactly.

The tower is the complete uniform- $e=2$  ancestry of the Mersenne numbers — the slowest exact approach to maximal fuel — and  $\lambda_d$  detects membership in its level- $d$  suffix pattern. The congruence domain is the order of 2 modulo  $3^d$ , and the carry-free normal form is an instance of the lifting-the-exponent lemma.

Section 8 turns from potentials to congruence case analysis. After the burn,  $2^n - 1$  reaches the repunit  $a_n = (3^n - 1)/2$ , and a natural attack on its stopping time scores the orbit of  $a_n$  block by block and splits on residues of  $n$ . Each such case is an exact finite computation (Proposition 8.4), but the method cannot cover all  $n$ :

*Theorem E* (Theorem 8.7). On  $n = 64k + 17$  with  $k$  odd, the 2-adic index  $k^*$  defined by  $3^{64k^*+20} = -367$  sends the orbit of  $a_n$  into the cycle  $-7 \mapsto -5 \mapsto -7$ , every block of which has

negative score. Integers  $k \equiv k^* \pmod{2^J}$  follow it for  $\lfloor (J+2)/3 \rfloor$  blocks, and no finite set of certified residue classes covers all odd  $k$ .

## Method and provenance

The constraint “ $\Phi$  nonincreasing” restricted to any coordinate window is a difference-constraint system on the values of  $g$ ; it is infeasible if and only if the coordinate graph of real orbit steps contains a directed cycle whose product of value ratios exceeds 1 (Section 7). Mining such cycles by Bellman–Ford over odd  $x \leq 10^6$  produces startlingly short certificates — e.g. the pair  $57 \mapsto 43$ ,  $27 \mapsto 41$  already refutes all  $g(x \bmod 16, \tau)$  — and inspection of the mined witnesses revealed that they shadow the  $-5$  cycle, which led to the uniform theorems above. We record this provenance because the historical route of this project ran in the opposite direction: Theorems A–C were first proved for subclasses by designed families (a Mersenne “burn” that forces linear growth of  $g$  along a frozen coordinate line, against a “recharge” family that caps it), a method that yields an exact price list — each level of suffix detection buys a would-be potential exactly  $\log_2 \frac{4}{3}$  of budget against an unbounded liability — but is dominated, as a source of impossibility statements, by the two-integer shadow certificates. The family analysis survives in the repository accompanying this paper and in the structural Theorem D.

## The boundary

A coordinate defeats the shadow only by separating  $x$  from  $f^{2a}(x)$  for every good Dirichlet approximation  $a$ : since the two agree on all low-order bits, fuel, suffix patterns, and (for suitable  $a$ ) leading digits, the surviving one-step class reads the *quantized full logarithm*  $\lfloor 2^j \log_2 x \rfloor$ . There a closed certificate needs total gain in  $(0, 2^{-j})$ , which pure shadows cannot supply; and as  $j \rightarrow \infty$  the class approaches arbitrary corrections, for which the existence of a monotone potential is conjecture-adjacent. Section 9 makes this precise.

## Relation to the literature

The parity-vector structure underlying the heuristic goes back to Terras [12] and Everett [3]; density results culminate in [4] and, by different methods, in Tao’s almost-all theorem [11]. Cycle exclusions rest on transcendence bounds [10, 9] and computation [2]. Backward (predecessor) structure of the Collatz graph is developed at the level of density and measure by Wirsching [13]; the tower of Theorem D is a distinguished explicit chain in that graph, and its closed form, congruence domain, and 2-adic freezing appear to be new. Forward identities of burn type, attached to the rational cycles at  $-1$ ,  $-5$ ,  $-17$ , appear in Andaloro [1] — the shadows of Section 3 are those identities viewed through the certificate lens. A search of the literature, including the annotated bibliographies [7, 8], located no prior impossibility theorems for concrete classes of one-step potentials; we would welcome references.

## 2 Preliminaries

Odd integers are written  $x$ ;  $v_p$  is the  $p$ -adic valuation;  $\tau(x) = v_2(x+1)$ ; congruential inverses are taken in  $(\mathbb{Z}/2^m\mathbb{Z})^\times$ . Two standard facts are used.

**Lemma 2.1** (LTE, special case). *For even  $n$ ,  $v_3(2^n - 1) = 1 + v_3(n)$ ; in particular  $v_3(2^{2 \cdot 3^{d-1}} - 1) = d$ .*

**Lemma 2.2.**  $2$  is a primitive root modulo  $3^d$  for every  $d \geq 1$ ; hence  $\text{ord}_{3^d}(2) = \varphi(3^d) = 2 \cdot 3^{d-1}$ .

*Proof.*  $2$  generates  $(\mathbb{Z}/9\mathbb{Z})^\times$  (its order divides 6 and is not 1, 2, 3), and a primitive root mod 9 is a primitive root mod  $3^d$  for all  $d$ .  $\square$

**Lemma 2.3** (fuse). *If  $\tau(x) = t \geq 2$  then  $v_2(3x+1) = 1$ ,  $f(x) = \frac{3x+1}{2}$ ,  $\tau(f(x)) = t-1$ , and  $f(x)/x = \frac{3}{2} + \frac{1}{2x}$ .*

*Proof.* Write  $x+1 = 2^t u$ ,  $u$  odd. Then  $3x+1 = 2(3 \cdot 2^{t-1}u - 1)$  with odd bracket since  $t \geq 2$ , and  $f(x)+1 = 3 \cdot 2^{t-1}u$ .  $\square$

*Remark 2.4.* The iterated form of Lemma 2.3 — that  $2^t u - 1$  ( $u$  odd) iterates in  $t$  odd steps to  $3^t u - 1$ , the *burn* — is classical; it is recorded by Andaloro [1] (cf. the annotation in [8]).

Finally we define the detectors. For  $d \geq 1$  write  $\text{ord}_d = 2 \cdot 3^{d-1}$  and, for  $M \equiv 1 \pmod{\text{ord}_d}$ ,  $M > 1$ ,

$$w_d(M) = \frac{2^{M+2d} - 2^{2d+1} + 3^d}{3^d} = 1 + 2^{2d+1} \frac{2^{M-1} - 1}{3^d}, \quad (1)$$

an odd integer by Lemma 2.2 (Theorem 6.1 develops its structure; nothing beyond the formula and Lemma 2.6 below is needed before Section 6).

**Definition 2.5.** For  $d \geq 1$  let  $\beta_d(s)$  be the bit length of  $T_s^{(d)} := w_d(1 + \text{ord}_d s)$  and define the *level- $d$  detector*

$$\lambda_d(x) = \max\{s \geq 1 : x \equiv T_s^{(d)} \pmod{2^{\beta_d(s)+1}}\} \in \mathbb{Z}_{\geq 1} \cup \{\perp\},$$

the maximum of the empty set being  $\perp$ .

**Lemma 2.6.** *Every template satisfies  $T_s^{(d)} \equiv 9 \pmod{32}$  if  $d = 1$  and  $T_s^{(d)} \equiv 1 \pmod{32}$  if  $d \geq 2$ . Consequently  $\lambda_d(y) = \perp$  for every  $d$  whenever  $y \not\equiv 1, 9 \pmod{32}$ .*

*Proof.* For  $d \geq 2$ ,  $2^{2d+1} \equiv 0 \pmod{32}$  in (1). For  $d = 1$ :  $M$  odd,  $M \geq 3$ , so  $2^{M-1} \equiv 0 \pmod{4}$ , whence  $(2^{M-1} - 1)/3 \equiv (-1) \cdot 3^{-1} \equiv 1 \pmod{4}$  and  $w_1(M) \equiv 1 + 8 \equiv 9 \pmod{32}$ . Matching any template forces the mod-32 congruence, since every matching modulus  $2^{\beta_d(s)+1}$  contains 32 ( $\beta_d(s) \geq 4$ ).  $\square$

### 3 The master theorem: shadowing the $-5$ cycle

The map has the rational cycle  $-5 \mapsto -7 \mapsto -5$ :  $f(-5) = -7$ ,  $f(-7) = -5$ , with multiplier  $3^2/2^3 = \frac{9}{8} > 1$ . Expanding cycles live at negative integers (the cycle equation gives  $x = c/(2^E - 3^K)$  with  $2^E < 3^K$ ), and their positive shadows gain value while walking the cycle's coordinate itinerary.

**Lemma 3.1** (shadow period). *Let  $N \geq 8$ ,  $w$  odd, and  $x = 2^N w - 5$ . Then*

$$v_2(3x+1) = 1, \quad f(x) = 3 \cdot 2^{N-1}w - 7, \quad v_2(3f(x)+1) = 2, \quad f^2(x) = 9 \cdot 2^{N-3}w - 5,$$

*so that  $8f^2(x) = 9x+5$  and  $f^2(x)/x = \frac{9}{8} + \frac{5}{8x}$ . Moreover  $\tau(x) = \tau(f^2(x)) = 2$  and  $\tau(f(x)) = 1$ ;  $x \equiv f^2(x) \equiv -5$  and  $f(x) \equiv -7 \pmod{2^{N-3}}$ ; and  $\lambda_d(x) = \lambda_d(f(x)) = \lambda_d(f^2(x)) = \perp$  for every  $d$ .*

*Proof.* Direct computation:  $3x+1 = 2(3 \cdot 2^{N-1}w-7)$  with odd bracket;  $3f(x)+1 = 4(9 \cdot 2^{N-3}w-5)$  with odd bracket;  $8f^2(x) = 9 \cdot 2^Nw - 40 = 9x + 5$ . Fuel:  $x+1 = 4(2^{N-2}w-1)$ ;  $f(x)+1 = 2 \cdot 3(2^{N-2}w-1)$  with odd second factor;  $f^2(x)+1 = 4(9 \cdot 2^{N-5}w-1)$  (odd bracket,  $N \geq 6$ ). Residues are immediate. Detectors:  $x \equiv f^2(x) \equiv 27$  and  $f(x) \equiv 25 \pmod{32}$  (using  $N-3 \geq 5$ ), and  $27, 25 \notin \{1, 9\}$ ; apply Lemma 2.6.  $\square$

**Theorem 3.2** (master no-go). *Let  $m, d \geq 0$  and let  $g$  be any real-valued function of  $(x \bmod 2^m, \tau(x), \text{len}(x), \lambda_1(x), \dots, \lambda_d(x))$ . Then  $\Phi(x) = \log_2 x + g(\dots)$  is not nonincreasing along  $f$  on odd  $x > 1$ .*

*Proof.* Take  $N \geq \max(m+3, 8)$ ,  $k \geq 3$ , and  $w = 2^{k-1} + 1$  in Lemma 3.1. Then additionally  $\text{len}(x) = \text{len}(f^2(x)) = N+k$ , since  $w$  places  $x$  in the lower part of its dyadic window:  $9(2^k+2) < 2^{k+4}$ . Hence  $x$  and  $f^2(x)$  have identical coordinate vectors while  $f^2(x) > x$ , and summing the two monotonicity constraints at  $x$  and  $f(x)$  gives  $0 \leq -\log_2(f^2(x)/x) < -\log_2 \frac{9}{8} < 0$ .  $\square$

The smallest member ( $N = 8, k = 3$ ) is  $1275 \mapsto 1913 \mapsto 1435$ , with  $8 \cdot 1435 = 9 \cdot 1275 + 5$  and ratio product  $\frac{287}{255}$ : two integers refute every  $g$  on the full coordinate algebra at  $m \leq 5$ .

*Remark 3.3* (general principle). Nothing is special about  $-5$ : every expanding rational cycle of the map — every solution of the cycle equation with  $2^E < 3^K$ , e.g. the  $K = 7$  cycle at  $-17$  with multiplier  $3^7/2^{11}$  — shadows at positive integers  $\equiv (\text{cycle member}) \bmod 2^N$  into a closed coordinate itinerary of value gain  $3^K/2^E > 1$ , refuting every correction that is 2-adically local along it. Expanding negative cycles are certificate factories against local potentials.

## 4 Iterating the shadow: leading digits

A coordinate can defeat a single shadow period only by separating  $x$  from  $f^2(x) = \frac{9}{8}x + \frac{5}{8}$ ; the leading digits do (the ratio  $\frac{9}{8}$  shifts them). Iteration removes this refuge.

**Theorem 4.1** (leading digits do not survive). *For every  $m, j, d \geq 0$  and every  $g$ , the potential  $\log_2 x + g(x \bmod 2^m, \tau(x), \text{LD}_j(x), \lambda_1(x), \dots, \lambda_d(x))$  is not nonincreasing along  $f$  on odd  $x > 1$ .*

*Proof.* For  $x \equiv -5 \pmod{2^{3a+c}}$ ,  $c \geq 5$ , Lemma 3.1 applies  $a$  times (each period consumes three bits of shadow depth), the itinerary alternating residues  $-5, -7$  with fuel  $2, 1$  and every detector  $\perp$  throughout, giving the exact closed form

$$f^{2a}(x) = \frac{9^a x + 5(9^a - 8^a)}{8^a} = \left(\frac{9}{8}\right)^a x (1 + \delta), \quad 0 < \delta < \frac{5}{x}.$$

Now  $\beta = \log_2 \frac{9}{8}$  is irrational ( $2^{p+3q} = 3^{2q}$  is impossible:  $v_3$  of the sides differ), so by Dirichlet there is  $a \geq 1$  with  $\|a\beta\| < 2^{-j-3}$  (distance to the nearest integer  $P$ ; note  $a\beta > 0$  regardless of the side). Choose  $N = 3a + \max(m, 5) + j + 8$  and a free odd multiplier  $w$  placing the fractional part of  $\log_2 x$ ,  $x = 2^N w - 5$ , at distance  $> 2^{-j-2}$  from every breakpoint of  $\text{LD}_j$  (possible since  $\log_2(2^N w - 5) \approx N + \log_2 w$  and  $\{\log_2 w\}$  is dense), with  $N$  large enough that  $\log_2(1+\delta) < 2^{-j-3}$ . Then  $\log_2 f^{2a}(x) = \log_2 x + P \pm \|a\beta\| + \log_2(1+\delta)$ , so  $\text{LD}_j$  closes; all other coordinates close by the itinerary; and telescoping monotonicity over the  $2a$ -step walk gives  $0 \geq a\beta + \log_2(1+\delta) > 0$ .  $\square$

A concrete instance ( $j = 4$ ):  $a = 53$  gives  $53\beta = 9.0060\dots$ ; the verifier realizes the full 106-step walk in exact integers at depth  $N = 177$ , with top-4 bits  $1011 \rightarrow 1011$ , all coordinates frozen, and value gain  $2^{9.006}$ .

## 5 Bounded corrections

**Theorem 5.1** (no bounded correction). *Let  $G : 2\mathbb{Z} + 1 \rightarrow \mathbb{R}$  be any bounded function. Then  $\log_2 x + G(x)$  is not nonincreasing along  $f$  on odd  $x > 1$ .*

*Proof.* Let  $\Omega = \sup G - \inf G$ . Telescoping monotonicity over the burn of  $2^M - 1$  (Lemma 2.3 iterated:  $M-1$  steps, each of ratio  $> \frac{3}{2}$ ) forces  $(M-1) \log_2 \frac{3}{2} \leq \Omega$ , false for  $M > 1 + \Omega / \log_2 \frac{3}{2}$ .  $\square$

Theorems A–C partition the correction landscape: bounded corrections fail on the burn alone; unbounded corrections through residues, fuel, length, leading digits, and suffix detectors of every depth fail on the shadow. Section 9 describes what is left.

## 6 The Mersenne ancestry tower

We now develop the structure behind the detectors. Define  $w_0(M) = 2^M - 1$  and  $w_d(M)$  as in (1).

**Theorem 6.1.** *For  $d \geq 1$  and  $M \equiv 1 \pmod{\text{ord}_d}$ ,  $M > 1$ :*

- (a) (Domain.)  $3^d$  divides  $2^{M+2d} - 2^{2d+1} + 3^d$  iff  $M \equiv 1 \pmod{\text{ord}_d}$ .
- (b) (Form.)  $w_d(M)$  is an odd integer  $> 1$  with  $\tau(w_d(M)) = 1$ .
- (c) (Descent.)  $3w_d(M) + 1 = 4w_{d-1}(M)$ ; hence  $v_2(3w_d(M) + 1) = 2$ ,  $f(w_d(M)) = w_{d-1}(M)$ , and  $f^d(w_d(M)) = 2^M - 1$ , each step of value ratio  $\frac{3}{4} + \frac{1}{4w_d} \in (\frac{3}{4}, 1)$ .
- (d) (Normal form.)  $B_d := (2^{\text{ord}_d} - 1)/3^d$  is an integer in  $(0, 2^{\text{ord}_d})$  by Lemma 2.1, and for  $M = 1 + \text{ord}_d s$ ,

$$w_d(M) = 1 + 2^{2d+1} \cdot [\text{the } \text{ord}_d\text{-bit block of } B_d]^s,$$

the bracket denoting  $s$  carry-free repetitions: an exactly periodic binary word of period  $2 \cdot 3^{d-1}$ .

- (e) (Freeze and separation.)  $w_d(M) \equiv \Lambda_d := 1 - 2^{2d+1}3^{-d} \pmod{2^{M-1}}$ ; and for  $d \neq d'$ ,  $v_2(\Lambda_d - \Lambda_{d'}) = 2 \min(d, d') + 1$ .

*Proof.* (a) Modulo  $3^d$  the numerator is  $2^{2d}(2^M - 2)$ , so divisibility is  $2^{M-1} \equiv 1 \pmod{3^d}$ ; apply Lemma 2.2. (b) The second form in (1) is 1 plus an even number, and  $w_d + 1 = 2(1 + 2^{2d}q)$  with  $2^{2d}q$  even. (c) Clearing  $3^{d-1}$ , the identity reduces to  $3^d + 3^{d-1} = 4 \cdot 3^{d-1}$ ;  $w_{d-1}$  is odd, so  $v_2 = 2$  exactly; iterate. (d) With  $M-1 = \text{ord}_d s$ ,  $(2^{M-1} - 1)/3^d = B_d \sum_{i=0}^{s-1} 2^{\text{ord}_d i}$ , a base- $2^{\text{ord}_d}$  number all of whose digits equal  $B_d < 2^{\text{ord}_d}$ : no carries; the  $+1$  and the shift by  $2^{2d+1}$  cannot interact. (e)  $3^d \cdot \frac{2^{M-1}-1}{3^d} \equiv -1 \pmod{2^{M-1}}$ ; and for  $d > d'$ ,  $\Lambda_d - \Lambda_{d'} = 2^{2d'+1}3^{-d}(3^{d-d'} - 2^{2(d-d')})$  with odd bracket.  $\square$

Levels 0, 1, 2 are the Mersennes, the family  $x'_M = (2^{M+2} - 5)/3 = (10)^{(M-3)/2}1001$  with  $f(x'_M) = 2^M - 1$ , and  $(111000)^k 01$ ; the tower is the complete uniform- $e=2$  ancestry of maximal fuel, and every tower member lies on the residue class  $8y + 1$ . Composing the tower with the burn (Remark 2.4) yields, for each  $(d, M)$ , an exactly known orbit segment of length  $d + M + 1$  terminating on the base-3 repunit  $(3^M - 1)/2$ .

**Lemma 6.2.** (1)  $\lambda_d(T_s^{(d)}) = s$ : the detector reads its own tower index, unboundedly. (2) For  $d \neq d'$ , with  $\nu = 2 \min(d, d') + 1$ : for all  $M > \nu + 1$  in the level- $d$  progression,  $\lambda_{d'}(w_d(M)) \leq \nu / \text{ord}_{d'}$ .

*Proof.* (1) At index  $s$  the congruence is an equality that holds; at  $s' > s$  the modulus exceeds both operands and templates strictly increase. (2) Suppose  $w_d(M)$  matches  $T_s^{(d')}$  with  $M' = 1 + \text{ord}_{d'} s$  and, toward a contradiction,  $M' - 1 \geq \nu + 1$ . The matching modulus refines  $2^{\nu+1}$  (as  $\beta_{d'}(s) \geq M'$ ), so Theorem 6.1(e) on both sides forces  $\Lambda_d \equiv \Lambda_{d'} \pmod{2^{\nu+1}}$ , contradicting  $v_2(\Lambda_d - \Lambda_{d'}) = \nu$ . Hence  $\text{ord}_{d'} s \leq \nu$ . (Small-index templates have not converged to  $\Lambda_{d'}$ ; this affects which small values occur, not the bound.)  $\square$

*Remark 6.3* (the family method and the price list). Historically, Theorems A–C were first obtained for subclasses by a family method that yields quantitative structure the shadow does not. The *burn* family  $3^j 2^t - 1$  ascends at ratio  $> \frac{3}{2}$  per step with frozen residue  $-1$  and  $\lambda \equiv \perp$ , forcing  $g$  to grow at slope  $\log_2 \frac{3}{2}$  along a coordinate line; the tower’s  $e=2$  chains cap the same values from above, each level costing one  $\log_2 \frac{4}{3}$ . The resulting price list — each detector level buys a would-be potential exactly  $\log_2 \frac{4}{3}$  of budget against an unbounded liability — explains *why* suffix information fails, while the shadow certificates prove *that* it fails. The full family proofs are retained in the repository.

## 7 Certificates from raw orbits

The constraint system “ $\Phi$  nonincreasing”, restricted to any coordinate window, is a system of difference constraints  $g(v) - g(u) \leq -\log_2(f(x)/x)$  over observed steps with  $u = \text{coords}(x)$ ,  $v = \text{coords}(f(x))$ ; it is infeasible iff the coordinate graph contains a directed cycle whose product of value ratios exceeds 1, and any such cycle is verifiable in exact rational arithmetic. Mining by Bellman–Ford over odd  $x \leq 10^6$  yields short certificates in every window we tested: the pair  $57 \mapsto 43$ ,  $27 \mapsto 41$  closes a 2-cycle in  $(x \bmod 16, \tau)$ -space with product  $\frac{1763}{1539} > 1$ , refuting Theorem 3.2’s class at  $m \leq 4$  in two lines; a 5-cycle through  $41 \mapsto 31 = 2^5 - 1$  does the same with the level-1 detector added. Inspection of mined witnesses at larger moduli revealed the  $-5, -7$  residue pattern of Section 3. This pipeline is logically independent of every lemma in this paper — it uses raw orbit steps only — and would have refuted the theorems had a valid correction existed on the tested windows.

## 8 Congruence certificates on the Mersenne spine

The preceding sections concern potentials. This one concerns a second method, case analysis on residue classes, applied to one explicit family of trajectories. By Lemma 2.3,  $f^j(2^n - 1) = 2^{n-j} 3^j - 1$  for  $0 \leq j < n$ . For odd  $n \geq 3$  the  $n$ -th step divides  $2(3^n - 1)$  by  $2^{1+v_2(3^n-1)} = 4$  (LTE), so

$$f^n(2^n - 1) = a_n := \frac{3^n - 1}{2},$$

the base-3 repunit, and the orbit stays above  $2^n - 1$  on the way. The stopping time of  $2^n - 1$  (first passage below the start) is therefore  $n + \sigma(a_n)$ , where  $\sigma(a_n)$  is the first passage of the orbit of  $a_n$  below  $2^n - 1$ . We know of no proof that  $\sigma(a_n)$  is finite for all odd  $n$ , let alone  $O(n)$ . A natural route scores the orbit of  $a_n$  block by block and case-splits on residues of  $n$ . We show

that such case splits are exact decision procedures on each residue class (Proposition 8.4), but that no finite collection of them can succeed (Theorem 8.7). The obstruction is again the cycle  $-5 \mapsto -7 \mapsto -5$ .

## Blocks and scores

Write  $T(x) = x/2$  for even  $x$  and  $T(x) = (3x + 1)/2$  for odd  $x$ , and read the orbit of  $x \in \mathbb{Z}_2$  under  $T$  as a parity word in the letters **O** (odd) and **E** (even).

**Definition 8.1.** Let  $x$  be odd with  $\tau(x) = 1$ , so  $e = v_2(3x + 1) \geq 2$ , and let  $y = f(x)$ . The *block* at  $x$  consists of the step  $x \mapsto y$  followed by the  $\tau(y) - 1$  steps of Lemma 2.3, which end at the next state with  $\tau = 1$ . Its *score* is

$$\Delta = 11(e - 1) - 9\tau(y).$$

In the parity word a block reads  $\mathbf{OE}^{e-1}\mathbf{O}^{\tau(y)-1}$ . It has  $e - 1$  even and  $\tau(y)$  odd letters, so  $\Delta = 11 \cdot \#\mathbf{E} - 9 \cdot \#\mathbf{O}$ . A run of blocks with nonnegative total score has at most  $11/20$  of its letters odd. Since  $3^{11} < 2^{20}$ , such a run contracts. Every block begins with the factor  $\mathbf{OE}$ , and  $\mathbf{OE}$  occurs nowhere else. Hence:

**Lemma 8.2.** *The scores  $\Delta_1, \dots, \Delta_B$  of the first  $B$  blocks from  $x$  are determined by any prefix of the parity word of  $x$  that contains the first  $B + 1$  occurrences of  $\mathbf{OE}$ .*

We use the subfamily  $n = 64k + 17$ . Here  $v_2(3^{n+1} - 1) = 3$  (LTE), so  $f(a_n) = x_1 := (3^{n+1} - 1)/8$ . Since  $3^{18} \equiv 9 \pmod{32}$  and  $3^{64} \equiv 1 \pmod{256}$ ,  $x_1 \equiv 1 \pmod{4}$  and  $\tau(x_1) = 1$ . Let  $\Delta_1, \Delta_2, \dots$  be the scores of the successive blocks from  $x_1$ . For odd  $k$  one has  $\Delta_1 = -16$  (Proposition 8.4 applied to the class  $k \equiv 1 \pmod{2}$ ). The score programme asks for  $\Delta_2 + \Delta_3 + \dots \geq 16$  summed over a window of  $6n$  shortcut steps. We call a class of  $k$  *closed at  $B$*  if  $\Delta_2 + \dots + \Delta_B \geq 16$  on the whole class. This prefix condition does not imply the window condition, since later blocks score  $-7, -16, \dots$  whenever  $\tau(y) \geq 2$ . Theorem 8.7 shows that even the prefix condition cannot be certified on all odd  $k$ .

## Exact decision on a residue class

**Lemma 8.3.** *For  $P \geq 6$ ,  $x_1 \bmod 2^P$  depends only on  $k \bmod 2^{P-5}$ . Moreover, if  $x \equiv x' \pmod{2^p}$  in  $\mathbb{Z}_2$ , then the parity words of  $x$  and  $x'$  agree in their first  $p$  letters.*

*Proof.* The order of 3 modulo  $2^{P+3}$  is  $2^{P+1}$ , so  $3^{64k+18} \bmod 2^{P+3}$  depends on  $64k \bmod 2^{P+1}$ . The second statement is Terras's observation [12]:  $T^i(x) \equiv T^i(x') \pmod{2^{p-i}}$  for  $i \leq p$ , by induction.  $\square$

**Proposition 8.4** (exact class decision). *Fix a class  $k \equiv r \pmod{2^j}$ . The first  $j + 5$  letters of the parity word of  $x_1$  are the same for every  $k$  in the class, including every 2-adic  $k$ . If they contain  $B + 1$  occurrences of  $\mathbf{OE}$ , then  $(\Delta_1, \dots, \Delta_B)$  is constant on the class. Otherwise split the class into its two children modulo  $2^{j+1}$  and repeat. A statement about  $(\Delta_1, \dots, \Delta_B)$  on a class is proved when it holds on every leaf of this tree, and refuted, with an explicit integer counterexample, when it fails on some leaf.*

*Proof.* Lemmas 8.3 and 8.2.  $\square$

For example, single class evaluations give

| $k$ class          | $(\Delta_2, \Delta_3, \Delta_4)$ | sum |
|--------------------|----------------------------------|-----|
| $3 \bmod 2^8$      | $(2, 13, 2)$                     | 17  |
| $171 \bmod 2^8$    | $(2, 2, 13)$                     | 17  |
| $323 \bmod 2^9$    | $(2, 13, 13)$                    | 28  |
| $579 \bmod 2^{10}$ | $(2, 13, 4)$                     | 19  |

so these four classes are closed at  $B = 4$ .

*Remark 8.5.* A single representative is not a certificate. The integers  $k = 267$  and  $k = 41227$  lie in the same class modulo  $2^{13}$ . For  $k = 267$ ,  $\Delta_2 + \dots + \Delta_6 = 43$ . For  $k = 41227$  the same sum is 7. The class modulo  $2^{13}$  does not determine  $\Delta_6$ . Proposition 8.4 refines it, and the leaf containing 41227 has modulus  $2^{18}$ .

## The cycle defeats every finite certificate

**Lemma 8.6.** *The map  $k \mapsto x_1(k)$  extends to a homeomorphism from the odd 2-adic integers onto the ball  $9 + 64\mathbb{Z}_2$ .*

*Proof.* Write  $3^{64} = 1 + 2^8 u$  with  $u$  odd. Then  $z \mapsto (1 + 2^8 u)^z$  is a homeomorphism from  $\mathbb{Z}_2$  onto  $1 + 2^8 \mathbb{Z}_2$  that scales distances by exactly  $2^{-8}$ , and odd  $z$  go onto  $257 + 2^9 \mathbb{Z}_2$ . The affine map  $w \mapsto (3^{18} w - 1)/8$  takes  $257 + 2^9 \mathbb{Z}_2$  onto  $(3^{18} \cdot 257 - 1)/8 + 64\mathbb{Z}_2 = 9 + 64\mathbb{Z}_2$ , using  $3^{18} \cdot 257 \equiv 73 \pmod{512}$ .  $\square$

So every point of  $9 + 64\mathbb{Z}_2$  is  $x_1$  of some odd 2-adic  $k$ . One of them falls straight into the expanding cycle of Section 3.

**Theorem 8.7** (shadow of the  $-5$  cycle on the spine). *Let  $k^* \in \mathbb{Z}_2$  be defined by  $3^{64k^*+20} = -367$ . Then*

1.  $k^*$  is odd,  $x_1(k^*) = -47/9$ , and its blocks have  $\Delta_1 = -16$  and  $\Delta_j = -7$  for every  $j \geq 2$ ;
2. for every  $J \geq 1$  and every integer  $k \geq 0$  with  $k \equiv k^* \pmod{2^J}$ ,

$$\Delta_2 = \Delta_3 = \dots = \Delta_B = -7, \quad B = \left\lfloor \frac{J+2}{3} \right\rfloor;$$

3. no finite set of residue classes of  $k$ , each closed at some  $B$  by Proposition 8.4, contains every odd integer  $k$ .

*Proof.* (1) One has  $-367 \cdot 3^{-20} \equiv 257 \pmod{512}$ , so  $k^*$  exists and is odd by the proof of Lemma 8.6. Then  $x_1 = (3^{64k^*+18} - 1)/8 = (-367/9 - 1)/8 = -47/9$ . Next  $3x_1 + 1 = -44/3$ , so  $e = 2$  and  $y_1 = -11/3$  with  $\tau(y_1) = v_2(-8/3) = 3$ , giving  $\Delta_1 = 11 - 27 = -16$ . Two steps of Lemma 2.3 give  $-11/3 \mapsto -5 \mapsto -7$ . From  $-7$  every block is  $-7 \mapsto -5 \mapsto -7$  with  $e = 2$  and  $\tau(-5) = 2$ , so  $\Delta = 11 - 18 = -7$ .

(2) The parity word of  $-47/9$  is  $\text{OEEO}(\text{OEO})^\infty$ . Its  $(B+1)$ -st occurrence of  $\text{OE}$  ends at letter  $3B+3$ . If  $k \equiv k^* \pmod{2^J}$ , Lemma 8.3 gives agreement in the first  $J+5$  letters, which contain  $B+1$  occurrences of  $\text{OE}$  when  $3B+3 \leq J+5$ . Lemma 8.2 finishes.

(3) Suppose classes  $C_1, \dots, C_m$  cover the odd integers. Refine them to a common modulus  $2^J$ . Every odd residue modulo  $2^J$  is then covered, including that of  $k^*$ , so some  $C_i$  contains  $k^*$ . The certificate for  $C_i$  uses only letters common to every 2-adic member of  $C_i$  (Proposition 8.4), so it applies to  $k^*$  and gives  $\Delta_2 + \dots + \Delta_{B_i} \geq 16$ . By (1) that sum is  $-7(B_i - 1) < 16$ , a contradiction.  $\square$

The mechanism is the one behind Theorem 3.2. The cycle block  $-7 \mapsto -5 \mapsto -7$  has one even and two odd letters and value ratio  $3^2/2^3 = \frac{9}{8} > 1$ . Any linear score whose nonnegativity forces contraction must therefore score this block negatively. Residue classes of  $k$  cannot separate  $k$  from  $k^*$ , just as residues of  $x$  cannot separate  $x$  from its shadow. In (2), each extra 3 bits of modulus buy exactly one more block of shadowing. One checks that  $B$  is sharp for  $J \leq 120$ .

## How far certificates reach

Proposition 8.4 also measures the method. Among odd  $k$ , the Haar measure of the classes modulo  $2^J$  that are closed within ten blocks is

$$\frac{1013}{2048} \approx 0.495 \ (J = 14), \quad \frac{77177}{131072} \approx 0.589 \ (J = 18), \quad \frac{10687}{16384} \approx 0.652 \ (J = 22).$$

The rest is undetermined at that modulus, apart from a part of measure about  $2.5 \times 10^{-4}$  at  $J = 22$  that provably does not close within ten blocks. Each four extra bits close about 6% more. By Theorem 8.7(3) the measure never reaches 1 at any finite  $J$ .

*Remark 8.8.* Even a complete cover would yield only prefix inequalities. The window condition also needs the later blocks, and a lower bound for them is the same problem for the shifted orbit. The full score programme of the repository accompanying this paper, if completed, would give  $\sigma(a_n) \leq 6n$  and hence a stopping time of at most  $7n$  odd steps for every  $2^n - 1$  with  $n$  odd. That is a linear stopping-time bound for an explicit infinite family whose orbits first climb by a factor of about  $(3/2)^n$ , and we know of no such result.

## 9 The boundary

What survives Theorem 4.1 must see  $\log_2 x$  to a precision finer than every  $\|a\beta\|$  — integer and fractional parts jointly: the *quantized full logarithm*. There, a closed certificate needs total gain in  $(0, 2^{-j})$ , which pure shadows cannot supply ( $\beta > 2^{-j}$  for  $j \geq 3$ ); any certificate must mix gain and loss segments, a connectivity question in the coordinate graph, minable per window but with no uniform construction known.

*Question 9.1.* Is there a nonincreasing potential of the form  $\log_2 x + g(\lfloor 2^j \log_2 x \rfloor, \tau(x), x \bmod 2^m)$  for some  $j, m$ ? As  $j \rightarrow \infty$  the class approaches arbitrary corrections, for which the existence of a monotone potential is conjecture-adjacent; the one-step no-go program has plausibly reached the point where further progress and the conjecture itself interact.

*Question 9.2.* Do multi-step potentials (functions nonincreasing along  $f^k$  for fixed  $k$ , or path-dependent accountings) admit analogous obstructions? The shadow supplies candidate constraints, since its  $k$ -step behaviour is exactly known; note that any  $k$ -step potential must in particular survive  $k$  dividing  $2a$  along the iterated shadow.

We also note what the results do *not* say. Apart from the negative statements of Section 8 about one method on one family, nothing here bears on individual trajectories, densities of divergent orbits, or cycles. The theorems delimit a proof *method*, not the dynamics.

## 10 Computational verification

Every numbered claim is verified by scripts using exact integer and rational arithmetic (no floating point in any assertion): the shadow lemma over wide parameter ranges with detec-

tor evaluation at all nodes (`verify_shadow.py`); the iterated shadow, the concrete Dirichlet instance  $(m, j, a) = (6, 4, 53)$  at depth  $N = 177$ , and the irrationality arithmetic (`verify_leading_digit.py`); the tower in all parts, including both directions of the domain criterion and the carry-free normal form (`verify_tower.py`); the family-method subclasses (`verify_no_local_potential.py`, `verify_nlp2.py`); the independent certificate miner with exact cycle confirmation (`verify_nogo_certificate.py`, `verify_fuel_fraction.py`); and for Section 8, the ball lemma, the shadow index  $k^*$ , part (2) of Theorem 8.7 for  $J \leq 120$ , the four class evaluations, and the exact closure measures (`verify_residue_certificate_nogo.py --measure-22`), with the class decision procedure in `prove_block8_17_classes.py`. The scripts accompany the repository.

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